

Quantitative, qualitative and mixed research

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Content:

1. A definition for quantitative and qualitative
 2. The purpose of quantitative and qualitative research
 3. How to collect quantitative and qualitative data
 4. On the application of quantitative and qualitative research
 5. Mixed
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Research strategies belong to one of two possibilities, quantitative or qualitative. There are differences between the two particularly when determining how data will be collected, analysed and the eventual result. The researcher has to be well aware of these basic differences and make an intelligent choice for the design of the research and achieving the research objectives. The research aim and nature of the research question are the driving factors in electing quantitative or qualitative research.

In IT, quantitative research is the more frequent choice. This is due to the prevailing application type or design research and less dependency on collecting data from humans.

1. A definition for quantitative and qualitative

Quantitative research is explaining phenomena by collecting numerical data that are analysed using mathematically based methods (in particular statistics).

Qualitative research seeks to answer questions about why and how people behave in the way that they do. It provides in-depth information about human behaviour.

(Skillsyouneed, 2021)

Quantitative research involves the collection of numerical data. This data is analysed using mathematical formula or statistics. The results of this type of research is based on the proof of the research methods.

Qualitative research does not rely on numerical data. Data collection includes perceptions, opinions, and observations. This type of research aims to seek answers to why and how *things* happen or exist.

2. The purpose of quantitative and qualitative research

Quantitative research usually aims to test or confirm theories or assumptions. The numerical data is transformed into statistics that quantifies the defined variables under investigation. This results in generalizable facts stemming from a sample to a much larger population.

Qualitative research aims to conduct deep investigations and gather in-depth insights into topics that are not well understood. Qualitative research is generally exploratory in nature. It attempts to understand the underlying reasons, opinions and motivations.

3. How to collect quantitative and qualitative data

Quantitative data is numerical and can be used to measure *variables*. Quantitative data is structured and statistical and the results are objective and conclusive. Quantitative data is collected from experiments, surveys and secondary data (e.g. ITS reports of student assessment marks).

An example of quantitative data might be:

A survey of 300 1st year IT students and ask them questions such as: “on a scale from 1-5, how satisfied are you with provisioned computer laboratories?”

Statistical analysis can be performed on the data and draw conclusions such as: “on average students rated computer laboratories at 4.4”

Data collection techniques include closed-ended questions.

Qualitative data is non-numerical and unstructured. Collected data seeks to describe a topic more than measure it. Data is usually collected from focus groups, interviews, observations and is collected in the *natural setting*. There usually are no simulations or experiments.

An example of qualitative data might be:

Conduct in-depth interviews with 15 1st year IT students and ask them open-ended questions such as: “How satisfied are you with provisioned IT labs?”, “What is the most positive aspect of the IT labs?” and “What can be done to improve the IT labs?”

The responses are transcribed, coded and then patterns and trends are identified.

Data collection techniques may include interviews, case studies, expert opinions, focus groups, open-ended survey and observation.

4. On the application of quantitative and qualitative research

For most research either the quantitative or qualitative approach can be used. The choice is determined by the nature of the research. A simple rule-of-thumb for use is:

Use quantitative research if you want to confirm or test something (a theory or hypothesis)

Use qualitative research if you want to understand something (concepts, thoughts, experiences)

(Scribbr, 2021)

However, some research combines both quantitative and qualitative data. For example, a survey can contain very precise closed-ended questions and at the end include an open-ended question. This open-ended question provides a means to capture any missed responses for example an open-ended question could be added to the end of the survey:

“Should you have any other comments regarding the IT Labs please write them here”.

In summary, the table below shows the difference between quantitative and qualitative research (scribbr.com, 2020)

Quantitative research	Qualitative Research
Focuses on testing theories and hypotheses	Focuses on exploring ideas and formulating a theory or hypothesis
Analysed through math and statistical analysis	Analysed by summarizing, categorizing and interpreting
Mainly expressed in numbers, graphs and tables	Mainly expressed in words
Requires many respondents	Requires few respondents
Closed (multiple choice) questions	Open-ended questions
Key terms: testing, measurement, objectivity, replicability	Key terms: understanding, context, complexity, subjectivity

Mixed Research

Mixed research is a combination of qualitative and quantitative. It usually involves data collected in unstructured, qualitative format from interviews, open ended questions, social media, video and sound clips. This data is transcribed or pre-processed into a numerical format. For example, data pertaining to race groups Indian, White, Black, Coloured, Other can be transcribed into a structured numerical format Indian=1; White=2; Black=3; Coloured=4; Other=5. Statistical or mathematical analysis techniques can then be applied.

Mixed research provides scientific rigour to the analysis of qualitative data.

Tutorials

1. What is the difference between quantitative and qualitative research?
2. Briefly describe “mixed-method” research.
3. Write a research question for a quantitative study
4. Write a research question for a qualitative study
5. Apart from the techniques listed in this lecture note, write down another two each for quantitative and qualitative data collection.
6. Given the following study aims, state whether each study is qualitative or quantitative
 - a. Developing a random forest model for Long-term follow-up of patients with chronic myeloid leukaemia
 - b. A crash sensor array to reduce road accident fatalities
 - c. Developing a student centred CX university portal
 - d. Identifying the challenges of Agile adoption in small IT solutions organizations
 - e. Building the first bio-fluid CPU

References

Scribbr.com. 2020. Qualitative vs. quantitative research. Available: <https://www.scribbr.com/methodology/qualitative-quantitative-research/>. Accessed: 15 January 2021.

Skillsyouneed. 2021. Quantitative and Qualitative Research Methods. Available: <https://www.skillsyouneed.com/learn/quantitative-and-qualitative.html>. Accessed: 15 January 2021.

Useful urls

<https://learn.g2.com/qualitative-vs-quantitative-data>

https://www.orau.gov/cdcynergy/soc2web/Content/phase05/phase05_step03_deeper_qualitative_and_quantitative.htm

<https://www.surveymonkey.com/mp/quantitative-vs-qualitative-research/>